No candidate

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

Nov 2013

CL201 Surveying

Date: 21.11.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Write principal of plane table surveying and give its advantages over other surveying methods. [05]

(b) What is EDM? Write principal of EDM. [02]

Q - 2 (a) Describe various accessories required for plane table surveying and discuss its function. Also discuss the steps for setting up a plane table. [07]

(b) Prepare gale's traverse table for the following data. [07]

Line	Length	Corrected W.C.B
AB	110	110°
BC	80	170°
CD	95	250°
DA	160	350°

OR

Q - 2 (a) Draw the sketch of following, explain radiation method. [07]

i) Alidade ii) U-fork iii) Trough compass

(b) An incomplete traverse table is obtained as follows. Find out missing data. [07]

Line	Length	Bearing
AB	194.1	85° 30'
BC	201.2	15°
CD	165.4	285° 30'
DE	172.6	185° 30'
EA	?	?

Q - 3 (a) Give functions for following parts of theodolite. [06]

i) Foot screw ii) Wing nut iii) Upper clamp screw iv) lower tangent screw v) Verniers C and D vi) Trough compass

(b) Explain procedure to set out building with neat sketch. [05]

(c) Differentiate between electronic theodolite and total station. [03]

OR

Q - 3 (a) Explain theodolite traversing procedure with fast angle method. [06]

(b) Write procedure to set out bridge pier. [05]

(c) Give the use of following parts of total station. [03]

i) Electronic vial ii) Laser plummet iii) Guide light

SECTION - II

Q - 4 (a) Explain **[03]**

i) Fly leveling ii) Check leveling iii) Sounding

(b) Give four characteristics of contours with sketch. **[04]**

Q - 5 (a) A Road embankment is 8 m wide and 200m in length at the formation level, with a side slope of 1.5:1. The embankment has a rising gradient of 1 in 100m. The ground levels at every 50 m along the centre line are as follows. The formation level of zero chainage is 166m. Calculate volume of earth work with Trapezoidal and Prismoidal rule. **[07]**

Distance(m)	0	50	100	150	200
R.L (m)	164.5	165.2	166.8	167	167.2

(b) Enlist methods to set horizontal curve with chain and tape, explain field procedure of any one method in detail. **[07]**

OR

Q - 5 (a) The ground level along the centre line of a rod is given below: **[07]**

Chainage(m)	0	50	100	150	200	250	300
G.L.	218	216.5	216	216.75	217.30	217.95	215.65

It is proposed that the formation level of RL 215.00 should be kept constant of starting from the chainage zero the formation width of the rod is 7 m and the side slope 1:1. The ground is level transverse to the centre line.

(b) Two tangents intersect at chainage 1190 m. the angle of deflection is 36°. Calculate all data necessary for setting out a curve of radius 300m by deflection angle method. The peg interval may be taken as 30m. Prepare a setting out table when lest count of vernier is 20". Calculate data for field checking. **[07]**

Q - 6 (a) The following offsets were taken at 10 m interval from a survey line to an irregular boundary line. 2.50, 4.40, 6.60, 5.50, 7.40, 8.70, 7.80, 6.50, 4.30, 3.20 m. Calculate the area enclosed between the survey line, the irregular boundary line and the first and last offset by (1) The trapezoidal rule (2) Simpson's rule. **[05]**

(b) Explain different types of horizontal curves with neat sketch. **[05]**

(c) Write purposes of hydrographic surveying. **[04]**

OR

Q - 6 (a) The following offsets are taken from a survey line to a curved boundary line: **[05]**

Distance (m)	0	5	10	15	20	30	40	60	80
Offsets (m)	2.40	3.70	4.50	5.30	6	4.80	5.70	3.80	2.10

Find the area between the survey line, the curved boundary line and the first and last offset by (1) The trapezoidal rule (2) Simpson's rule.

(b) Draw neat sketch of circular and compound curve with its name and notation of different components. **[05]**

(c) What is a control point? Explain horizontal and vertical control in context of hydrographic survey. **[04]**

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